PROJECT STATEMENT 2

DATABASE MANAGEMENT SYSTEMS · COMPREHENSIVE PROJECT SUBMISSION

L&T EDUTECH PROJECT

INVENTORY MANAGEMENT SYSTEM

A Cloud-Hosted Relational Database Solution featuring Multi-Table Tracking, Foreign Key Integrity, Automated Inventory Analytics & a Live Web Dashboard.

LIVE PRODUCTION DEPLOYMENT

https://inventorymanagement-lake.vercel.app

Database Engine: MySQL 8.4 (DigitalOcean BLR · Aiven Cloud)
Backend Architecture: Python 3.14 · Flask Framework
Hosting Provider: Vercel Global Edge Network

Project ID: Statement 2
Security: SSL / TLS CA-Verified Connection
Submission Year: 2026

1. PROJECT OVERVIEW & OBJECTIVES

In contemporary retail and supply chain operations, accurate tracking of stock, reliable supplier information, and comprehensive audit trails of goods movement are critical. **Project Statement 2: Inventory Management System** provides an enterprise-ready database and application design engineered to handle warehouse logistics, real-time stock deductions, restock shipments, and defective returns.

1.1 Core Objectives

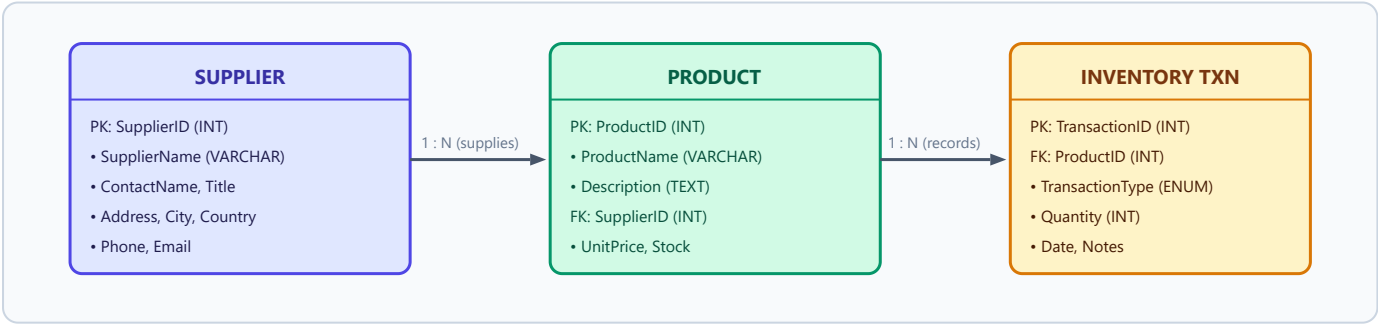
- **Supplier Relationship Management**: Centralize complete contact, geographic, and communication credentials for all vendor organizations.
- **Product Catalog Standardization**: Maintain comprehensive SKU records with detailed descriptions, supplier linkage, unit pricing, and real-time inventory counts.
- **Transactional Movement Auditing**: Record every inflow and outflow using strongly typed movement classifications (`Purchase` , `Sale` , `Return`) with timestamps and order notes.
- **Automated Business Intelligence**: Deliver instant SQL metrics for low-stock thresholds, inventory asset valuation, and supplier fulfillment performance.
- **Interactive Cloud Accessibility**: Eliminate static spreadsheets by integrating the database into a secure, responsive web portal hosted in the cloud.

1.2 Technology Architecture

Database Management System	**MySQL 8.4 Community Server** hosted on Aiven Cloud (Bangalore Region)
Connection Security	**Encrypted SSL / TLS** with custom Certificate Authority (CA) verification
Application Backend	**Python 3.14** with **Flask WSGI Framework** & `mysql-connector-python`
Frontend Client	**HTML5, Vanilla JavaScript, TailwindCSS, SVG Icons** (Light Minimalist Theme)
Cloud Web Deployment	**Vercel Serverless Platform** (inventorymanagement-lake.vercel.app)

2. ENTITY-RELATIONSHIP (ER) ARCHITECTURE

The system follows Third Normal Form (3NF) relational principles to eliminate data redundancy while maintaining relational integrity via Foreign Key cascade rules.



3. DATA DICTIONARY & TABLE DEFINITIONS

3.1 Supplier Table

FIELD NAME	DATA TYPE	CONSTRAINTS	FUNCTIONAL DESCRIPTION
`SupplierID`	INT	PK AUTO_INCREMENT	Unique surrogate identifier for vendor
`SupplierName`	VARCHAR(150)	NOT NULL	Registered corporate trade name
`ContactName`	VARCHAR(100)	Nullable	Principal account representative
`ContactTitle`	VARCHAR(100)	Nullable	Corporate designation (e.g. Sales Manager)
`Address`	VARCHAR(200)	Nullable	Warehouse / Headquarters street address
`City` , `Region`	VARCHAR(100)	Nullable	City & State / Territorial jurisdiction
`PostalCode`	VARCHAR(20)	Nullable	Postal PIN routing code
`Country`	VARCHAR(100)	Nullable	Operating Country
`Phone`	VARCHAR(30)	Nullable	Telephone number with dial prefix
`Email`	VARCHAR(150)	Nullable	Official correspondence email

3.2 Product Table

FIELD NAME	DATA TYPE	CONSTRAINTS	FUNCTIONAL DESCRIPTION
`ProductID`	INT	PK AUTO_INCREMENT	Unique stock-keeping unit (SKU) identifier
`ProductName`	VARCHAR(150)	NOT NULL	Marketable brand/item title
`Description`	TEXT	Nullable	Product technical specifications
`SupplierID`	INT	FK → Supplier	Source vendor ID (ON UPDATE CASCADE ON DELETE SET NULL)
`UnitPrice`	DECIMAL(10,2)	NOT NULL DEFAULT 0.00	Price per individual unit in INR (₹)
`QuantityInStock`	INT	NOT NULL DEFAULT 0	Current physical warehouse count

3.3 InventoryTransaction Table

FIELD NAME	DATA TYPE	CONSTRAINTS	FUNCTIONAL DESCRIPTION
`TransactionID`	INT	PK AUTO_INCREMENT	Unique sequential ledger ID
`ProductID`	INT	FK → Product	Target product affected (ON UPDATE CASCADE ON DELETE CASCADE)
`TransactionType`	ENUM	'Purchase', 'Sale', 'Return'	Classification of warehouse movement
`Quantity`	INT	NOT NULL	Volume of units moved in this transaction
`TransactionDate`	DATE	NOT NULL	Date of movement execution
`Notes`	TEXT	Nullable	Invoice numbers, order notes, audit remarks

4. SQL IMPLEMENTATION (DDL & SEED DATA)

4.1 Complete Schema Creation Script

```
-- Drop existing tables in reverse dependency order
SET FOREIGN_KEY_CHECKS = 0;
DROP TABLE IF EXISTS InventoryTransaction; DROP TABLE IF EXISTS Product; DROP TABLE IF EXISTS Supplier;
SET FOREIGN_KEY_CHECKS = 1;

CREATE TABLE Supplier (
```

```
    SupplierID   INT          PRIMARY KEY AUTO_INCREMENT,
    SupplierName VARCHAR(150) NOT NULL,
    ContactName  VARCHAR(100), ContactTitle VARCHAR(100),
    Address      VARCHAR(200), City VARCHAR(100), Region VARCHAR(100),
    PostalCode   VARCHAR(20),  Country VARCHAR(100), Phone VARCHAR(30), Email VARCHAR(150)
);

CREATE TABLE Product (
    ProductID       INT            PRIMARY KEY AUTO_INCREMENT,
    ProductName     VARCHAR(150)   NOT NULL,
    Description     TEXT,
    SupplierID      INT,
    UnitPrice       DECIMAL(10,2)  NOT NULL DEFAULT 0.00,
    QuantityInStock INT            NOT NULL DEFAULT 0,
    CONSTRAINT fk_product_supplier FOREIGN KEY (SupplierID) REFERENCES Supplier(SupplierID)
        ON UPDATE CASCADE ON DELETE SET NULL
);

CREATE TABLE InventoryTransaction (
    TransactionID   INT  PRIMARY KEY AUTO_INCREMENT,
    ProductID       INT  NOT NULL,
    TransactionType ENUM('Purchase', 'Sale', 'Return') NOT NULL,
    Quantity        INT  NOT NULL,
    TransactionDate DATE NOT NULL,
    Notes           TEXT,
    CONSTRAINT fk_transaction_product FOREIGN KEY (ProductID) REFERENCES Product(ProductID)
        ON UPDATE CASCADE ON DELETE CASCADE
);
```

5. CORE SQL QUERIES & EXECUTION RESULTS

5.1 Product Catalog Joined with Supplier Information

```
SELECT p.ProductID, p.ProductName, s.SupplierName, p.UnitPrice, p.QuantityInStock
FROM Product p LEFT JOIN Supplier s ON p.SupplierID = s.SupplierID ORDER BY p.ProductID;
```

ID	PRODUCT NAME	SUPPLIER ORGANIZATION	UNIT PRICE	IN STOCK
1	Laptop Pro 15	TechParts Inc.	₹75,000.00	50 units
2	Wireless Mouse	Global Supplies	₹850.00	200 units
3	USB-C Hub	QuickStock Ltd.	₹2,500.00	150 units
4	Mechanical Keyboard	TechParts Inc.	₹4,500.00	80 units
5	Monitor 27 inch	FastShip Co.	₹32,000.00	30 units
6	Webcam HD	Global Supplies	₹3,200.00	120 units
7	Desk Lamp LED	QuickStock Ltd.	₹750.00	300 units
8	Headset Pro	FastShip Co.	₹6,500.00	60 units

5.2 Stock Movement Summary (Purchased vs. Sold vs. Returned)

```
SELECT p.ProductName,
    COALESCE(SUM(CASE WHEN t.TransactionType = 'Purchase' THEN t.Quantity ELSE 0 END), 0) AS TotalPurchased,
    COALESCE(SUM(CASE WHEN t.TransactionType = 'Sale'     THEN t.Quantity ELSE 0 END), 0) AS TotalSold,
    COALESCE(SUM(CASE WHEN t.TransactionType = 'Return'   THEN t.Quantity ELSE 0 END), 0) AS TotalReturned,
    p.QuantityInStock AS CurrentStock
FROM Product p LEFT JOIN InventoryTransaction t ON p.ProductID = t.ProductID
GROUP BY p.ProductID, p.ProductName, p.QuantityInStock ORDER BY p.ProductName;
```

PRODUCT NAME	PURCHASED	SOLD	RETURNED	CURRENT STOCK BALANCE
Desk Lamp LED	150	70	0	300 units
Headset Pro	40	15	2	60 units
Laptop Pro 15	20	5	0	50 units
Mechanical Keyboard	50	10	0	80 units
Monitor 27 inch	15	3	0	30 units
USB-C Hub	80	20	0	150 units
Webcam HD	60	25	0	120 units
Wireless Mouse	100	30	5	200 units

5.3 Low-Stock Critical Alert Report (< 50 units threshold)

```
SELECT ProductID, ProductName, QuantityInStock, 'CRITICAL LOW STOCK' AS Status
FROM Product WHERE QuantityInStock < 50 ORDER BY QuantityInStock ASC;
```

SKU ID	PRODUCT DESCRIPTION	QUANTITY AVAILABLE	SYSTEM STATUS
5	Monitor 27 inch	30 units	CRITICAL LOW STOCK (Reorder Required)

5.4 Warehouse Financial Asset Valuation

```
SELECT p.ProductName, p.UnitPrice, p.QuantityInStock, ROUND(p.UnitPrice * p.QuantityInStock, 2) AS LineValuation
FROM Product p ORDER BY LineValuation DESC;
```

PRODUCT ITEM	UNIT PRICE	STOCK COUNT	ASSET VALUATION (₹)
Laptop Pro 15	₹75,000.00	50	₹37,50,000.00
Monitor 27 inch	₹32,000.00	30	₹9,60,000.00
Headset Pro	₹6,500.00	60	₹3,90,000.00
Webcam HD	₹3,200.00	120	₹3,84,000.00
USB-C Hub	₹2,500.00	150	₹3,75,000.00
Mechanical Keyboard	₹4,500.00	80	₹3,60,000.00
Desk Lamp LED	₹750.00	300	₹2,25,000.00
Wireless Mouse	₹850.00	200	₹1,70,000.00
TOTAL WAREHOUSE CAPITAL VALUATION			**₹66,14,000.00**

6. WEB DASHBOARD & CLOUD DEPLOYMENT

To eliminate spreadsheet fragmentation, a live reactive web dashboard was constructed with Python Flask and hosted serverless on Vercel, directly interfacing with the Aiven Cloud MySQL database over verified SSL/TLS.

> **Production Web Portal: https://inventorymanagement-lake.vercel.app**

6.1 Full CRUD Architecture

- **Create (POST)**: Modal dialogues allow adding new products (with supplier binding), vendors, and transactions without manual SQL queries.
- **Read (GET)**: Live reactive tabular feeds showing real-time stock balances, supplier contacts, and transaction logs.
- **Update (PUT & PATCH)**:
  - *Full Modal Edit*: Allows updating pricing, descriptions, and supplier assignments.
  - *Quick-Stock Stepper*: Inline `[-]` and `[+]` incrementers allowing warehouse workers to adjust unit counts in a single click.
- **Delete (DELETE)**: Double-confirmation modal alerts preventing accidental permanent removals.

6.2 Security & Integrity Features

- **SSL/TLS Certificate Authority Verification**: All connections between Vercel Serverless Functions and Aiven MySQL are encrypted using an embedded `ca.pem` certificate, guaranteeing zero plaintext transit.
- **SQL Injection Prevention**: All API queries utilize parameterized prepared statements (`%s` placeholders via `mysql.connector`).
- **Cross-Site Scripting (XSS) Sanitization**: Input strings are sanitized via utility escaping functions before rendering in the DOM.

7. CONCLUSION & SUBMISSION DELIVERABLES

Project Statement 2: Inventory Management System has been executed to completion. The relational design satisfies all functional constraints, maintains transactional fidelity across all goods movements, and delivers instant operational visibility through automated SQL queries and a live web application.

DELIVERABLE ITEM	FORMAT	STATUS	VERIFICATION DETAILS
Complete SQL Script	`.SQL`	**Verified**	`inventory_management.sql` (DDL, DML, 14 Queries)
Project Submission Report	`.PDF`	**Verified**	`Inventory_Management_System_Project_Report.pdf`
Database Data Dump	`.XLSX`	**Verified**	`Inventory_Management_Data.xlsx` (3 Sheets)
Live Web Application	HTTP / Cloud	**Live**	https://inventorymanagement-lake.vercel.app
Cloud MySQL Database	Aiven Cloud	**Running**	`mysql-321170c9-inventorymanagement-system.f.aivencloud.com:10525`